Sparse Impulsive-Noise Removal: Classical Detection and Repair versus Neural Speech Enhancement

Leona Wang | SID 540686380 | GitHub: zhi0708 | ELEC5305

Repository: https://github.com/zhi0708/elec5305-project--540686380-

Project site: https://zhi0708.github.io/elec5305-project--540686380-/

Abstract

Sparse impulsive interference presents a preservation problem: removing a short event can damage considerably more speech than the event itself. This project compares event-localised DSP with an official pretrained GTCRN enhancer using controlled insertions into a reference speech recording. An energy detector and a three-feature STFT detector drive local attenuation or linear interpolation. The experiment uses 90 development mixtures and 180 held-out mixtures, covering five event templates, three interference levels and three acoustic locations. Exact insertion intervals are available for every case. The STFT detector achieves event F1 0.362, compared with 0.080 for energy detection, while retaining low recall. GTCRN reduces mean event reconstruction error more strongly (5.87 dB), but introduces greater clean-region alteration than the STFT local methods. These results show a restoration-versus-preservation trade-off. On 1,728 mixtures from three held-out speakers, event F1 was 0.084 for the original STFT detector, 0.154 after retuning STFT, and 0.475 for the short-context detector.

1. Research question and motivation

The research question is: for sparse short-duration impulsive interference in speech, how does event-localised DSP detection and repair compare with a modern pretrained neural speech enhancer in event restoration, speech preservation and computational cost? A secondary question concerns interference overlapping an existing speech transient.

Predictive reconstruction [1], stationary-wavelet suppression [2] and HHT detection [3] provide historical context for specialised impulsive-noise processing. The current project implements neither the wavelet nor HHT algorithms. Recent sparse adaptive filtering [5] motivates examining disturbance sparsity and localisation: its author repository discusses sparse optimisation and the consequences of altering undisturbed samples. It is used as motivation rather than reproduced or claimed as a baseline. GTCRN [4] supplies an accessible lightweight learned comparator with official pretrained weights. The project contribution is the controlled experiment, transparent student-developed detector and separation of event error from collateral alteration.

2. Data and controlled corruption

The existing repository contains one 33.62-second, mono 8 kHz speech recording. It is OSR_us_000_0010_8k.wav from the Open Speech Repository, verified against a byte-identical public mirror. Its source and SHA-256 are recorded in data/speech_manifest.json. It is treated as a controlled reference rather than a certified noise-free corpus. Only the first 30 seconds are used. They form six non-overlapping five-second clips: clips 1-2 are development data and clips 3-6 are test data. The split separates time segments, not speakers. Upsampling to 16 kHz meets GTCRN's input requirements but does not restore missing bandwidth. All speech clips share one fixed scaling that limits the original recording peak to 0.15; mixtures and outputs are never independently normalised.

Two synthetic templates are seeded Gaussian bursts with Hann envelopes, lasting 5 and 20 ms. Three real templates are cropped from ESC-50 [6]: door knock (31.375 ms), mouse click (16.125 ms) and keyboard event (18.312 ms). A 2 ms RMS envelope locates the strongest peak; the crop expands while above 10% of peak, with a 75 ms limit on each side, 2 ms guards and short boundary fades. Source files, crop times, individual attribution and file hashes are provided. There is only one extracted template per real class, so this tests three examples rather than environmental-event diversity.

Each template is added separately at a pause, a steady voiced candidate and a speech-transient candidate. Locations are derived from the reference waveform: minimum smoothed energy identifies the pause; autocorrelation with an energy gate and low-flux preference identifies steady voiced speech; large positive spectral flux identifies the transient. These are acoustic proxies, not human-verified vowels or phoneme-aligned /p/, /t/ and /k/ releases. The reference is used for constructing controlled cases and evaluating outputs, never supplied to a tested detector.

Event levels are -6, 0 and +6 dB. Level means event RMS relative to the RMS of a local speech window extending 50 ms on each side of the inserted interval. In near silence, that reference has a floor at 10% of whole-clip RMS. An insertion mask covers the exact half-open interval [start, end), including its fades. The generator retains clean speech, added event, corrupted speech and the mask. All case intervals are saved in the manifest; scripts/export_case.py materialises any case. Float arrays are used for measurements to avoid clipping or quantisation. The grid contains 2 x 5 x 3 x 3 = 90 development mixtures and 4 x 5 x 3 x 3 = 180 test mixtures. These are controlled variants of a small reference, not 270 independent utterances.

3. Detection and local restoration

The analysis uses 8 ms windows and 2 ms hops, with a Hann window for spectra and a 256-point real FFT. Short frames favour transient timing; zero padding samples the spectrum more densely but does not increase physical frequency resolution. Short windows improve temporal localisation at the expense of frequency discrimination. Energy describes local intensity, spectral flux captures positive frame-to-frame spectral change, and bandwidth describes spectral spread. Suppression is evaluated separately from speech preservation; LPC reconstruction remains a possible extension.

The energy baseline compares a local energy rise against a threshold. Its rise is 10 log10((frame_energy + floor)/(local_median_energy + floor)). The median spans 101 frames, approximately 202 ms; the floor is 1% of the recording's 60th-percentile frame energy. This adaptive baseline can detect sudden increases but can mistake normal speech onsets for interference. Its runtime excludes all FFT calculations.

The STFT method combines three features: positive energy rise, positive spectral flux and spectral bandwidth. Flux is the sum of squared positive changes between consecutive L2-normalised

magnitude spectra. Its robust z-score uses the recording median and a scale equal to the maximum of 1.4826 times the median absolute deviation, 0.1 times its standard deviation, and a numerical floor. Bandwidth is the square root of the power-weighted frequency variance. The score is max(energy_rise,0)/6 + max(flux_z,0)/3 + bandwidth/1500. The coefficients are fixed heuristic scales, not learned from test labels. Their benefit is interpretability; no ablation is available to isolate each feature's contribution.

Candidate frames exceeding the threshold are mapped to sample intervals, including the frame's half-window support. Candidates longer than 150 ms are rejected. The same duration rule applies to both detectors. The development search considers seven energy thresholds and eight STFT thresholds. It maximises event F1 with false events on the two clean development clips counted 45 times, giving each clean clip the same weight as its 45 noisy variants. This makes preservation an explicit selection preference. The frozen thresholds are 9 dB energy rise and STFT score 5. The test set is not used to choose either threshold.

Attenuation applies a gain of 0.2 inside the predicted event, with up to 2 ms entry and exit ramps. Interpolation replaces the predicted samples with a straight line connecting the preceding and following samples. At file boundaries, the missing neighbour is zero. Both methods preserve the input exactly outside the predicted mask. This guarantee concerns the predicted interval; false detections can still alter genuinely clean speech. An oracle interpolation control uses the true mask solely to diagnose reconstruction limitations and is excluded from detector performance and runtime comparison.

4. Neural comparison and evaluation protocol

GTCRN uses the official DNS3 checkpoint, unmodified model architecture and a pinned source commit. No model is trained or fine-tuned. The adapter follows the official 512-sample FFT, 256-sample hop and square-root Hann analysis/synthesis convention. Modern PyTorch complex STFT tensors are converted into the model's real/imaginary input representation and converted back for iSTFT. Output length is explicitly matched to input length. There is no case-specific gain correction, delay fitting or reference-guided enhancement.

Event detection uses one-to-one matching of predicted and true intervals with positive overlap. TP, FP, FN, precision, recall and F1 are pooled over test cases. A partial overlap counts as an event hit, so sample-level metrics are also stored to reveal poor interval coverage. Onset error is the absolute start-time difference for matched events only; it does not summarise misses. Reports separate pause, steady voiced and transient locations, and retain event type and strength in per-case CSVs.

Event RMSE compares output with the clean reference inside the true insertion interval. Collateral RMSE uses every sample outside that true interval, thereby counting false-positive repair and any neural alteration. Event error reduction is 10 log10(noisy event MSE / output event MSE); positive values mean improvement. Whole-file SI-SDR and STOI [7] are secondary checks, not substitutes for local measurements. Table values are arithmetic means across cases; median RTF is reported separately. A mean of dB improvements is not equivalent to a dB ratio of mean errors. Clean-input preservation is evaluated over all four test clips (20 seconds) with no inserted events.

Both methods run on the same INTEL(R) XEON(R) PLATINUM 8573C CPU environment. Neural inference is CPU-only with one PyTorch thread. Model loading and one warm-up call are excluded. Local runtime includes detector computation and repair; neural runtime includes STFT, model and iSTFT. RTF is wall-clock processing time divided by five seconds of audio. It measures offline throughput, not causal latency: local median filtering and recording-wide feature normalisation use future context. No streaming implementation is claimed.

5. Measured results

5.1 Detection

Detector	Precision	Recall	F1	TP / FP / FN	Mean matched onset error (ms)
ENERGY	0.047	0.289	0.080	52 / 1062 / 128	3.77
STFT	0.521	0.278	0.362	50 / 46 / 130	2.76

The STFT detector markedly increases precision and overall F1 but does not improve recall. Both methods miss most of the weak events. The STFT method has 130 missed events out of 180; it is a conservative detector, not a reliable general impulse remover. Its sample-level recall is only 0.087, lower than event recall, because some detections overlap only a fraction of an inserted event. Inference from event F1 alone would therefore overstate useful localisation.

Acoustic location	Detector	Precision	Recall	F1
silence	energy	0.003	0.017	0.005
silence	stft	0.545	0.300	0.387
speech transient	energy	0.080	0.517	0.139
speech transient	stft	0.579	0.367	0.449
steady voiced	energy	0.052	0.333	0.091
steady voiced	stft	0.400	0.167	0.235

Figure 1 shows precision, recall and F1 against strength separately for these locations. Full grouped values also separate the five event templates. The strongest performance is template-dependent: the short synthetic click and real mouse click are detected more often than the cropped knock and keyboard event. This is consistent with the chosen spectral features favouring abrupt broad-spectrum change, but without an ablation the explanation remains a hypothesis. The transient condition is not uniquely hardest in this small experiment, so the experiment does not establish consonant-overlap conditions as the most difficult category.

5.2 Restoration, preservation and computation

Method	Event RMSE	Clean-region RMSE	Event error reduction (dB)	SI-SDR (dB)	STOI	Median RTF
Unprocessed	0.016776	0.000000	0.00	32.95	0.971	0.00000
Energy + interpolation	0.013971	0.004933	1.25	12.13	0.877	0.00030
STFT + attenuation	0.014943	0.000026	0.97	33.77	0.972	0.00183
STFT + interpolation	0.014994	0.000062	1.15	33.64	0.974	0.00183
GTCRN (DNS3 pretrained)	0.006374	0.002243	5.87	17.89	0.899	0.01649

GTCRN produces the lowest measured mean event RMSE and about 5.87 dB mean event error reduction, compared with 1.15 dB for STFT interpolation. However, STFT interpolation has far lower collateral RMSE (0.000062 versus 0.002243) and preserves global speech metrics more closely. GTCRN's mean whole-file STOI is lower than the unprocessed sparse-corruption input

despite better local event repair. This shows why event removal alone does not establish whole-speech benefit. It may also reflect domain mismatch: the reference originates at 8 kHz and is upsampled, whereas the model is a general 16 kHz enhancer. The experiment cannot isolate domain mismatch from learned processing effects.

The energy baseline has particularly high collateral error because its false events repair many legitimate speech intervals. Locality alone does not guarantee preservation if the detector is wrong. STFT attenuation and interpolation yield modest average improvement and a zero median improvement: at least half of cases receive no useful repair. The oracle control yields mean event RMSE 0.015971 and mean event error reduction 2.85 dB. Even perfect event location does not guarantee that a straight line reconstructs oscillatory speech. Its event RMSE can remain large on speech overlaps despite exactly zero collateral error.

Figure 2 displays event error against collateral error. Figure 3 uses the same 20 ms synthetic event waveform at the same absolute amplitude in all three locations; it is a separate demonstration from the local-RMS strength sweep. The fixed amplitude is set from the +6 dB steady-voiced example and is not rescaled in silence or at the transient. Waveform and spectrogram panels use identical inputs for all methods. Figure 5 shows the measured runtime trade-off. All measured median RTFs are below one, but this is an offline CPU result. GTCRN parameter tensors occupy 192,980 bytes; the checkpoint is 579,819 bytes. The model has 48,245 parameters including fixed ERB mappings, of which 23,669 are trainable. These are model-storage figures, not measured process peak memory; DSP stores no learned weights but does use waveform, frame and FFT work buffers.

5.3 Clean-input false alarms and preservation

Method	False events / 20 s	False events / minute	Mean clean RMSE	Modified clean samples
Energy + interpolation	24	72	0.00492323	3.5575%
STFT + interpolation	1	3	0.00000564	0.0403%
GTCRN (DNS3 pretrained)	N/A	N/A	0.00221101	100.0000%

The STFT detector produces one false event over the clean test clips, versus 24 for the energy baseline. Figure 4 shows that STFT false alarm in clip 6: a small waveform change coincides with a sharp normalised spectral-flux peak, rather than an unusually loud pulse. This explains how abrupt spectral evolution can cross the threshold even at low absolute energy. The waveform has not been phoneme-labelled, so the event is described as a legitimate reference-speech change, not identified as /t/ or /k/. The clean no-op test also shows that GTCRN changes almost all samples; the changes are numerically measurable, not evidence that every altered sample sounds worse. Audio examples enable informal inspection, but no formal listening study or perceptual preference claim is made.

6. Speaker-disjoint evaluation

6.1 Protocol and detector

The extension uses L2-ARCTIC v5.0 [8]. The first six manually annotated utterances in sorted order for ASI and HKK form the development set. The next eight utterance IDs for each of BWC, TLV and YDCK form the test set. Neither speakers nor prompt IDs cross the split. Waveforms are resampled from 44.1 kHz to 16 kHz and each reference is scaled once to a peak of 0.15. The reference, mixture and outputs then share that scale. In development, the original five event templates yield 540 mixtures. Testing yields 1,728 mixtures, including 648 using three additional ESC-50 recordings from fold 2. Clean controls total 120.0 seconds. These are controlled variants of 24 test utterances from three speakers.

The short-context detector measures first-difference energy in 2 ms windows. It compares this energy with the larger mean of the two flanks, approximately 20-60 ms away. This downweights rises that continue into surrounding speech. Its score combines positive flank contrast in dB (scaled by 6), positive local energy rise (scaled by 9), bandwidth (scaled by 1700 Hz), capped flux $\min(3 \times \text{flux}, 1)$, and an optional low-level penalty. Development selected contrast weight 0.75, low-level penalty weight 0.0, and threshold 4.0. A threshold-only STFT control selected 4.0 from the original eight-value grid. Both searches used the same development clips and maximised event F1 with each clean false alarm weighted by 45. Test mixtures and phoneme labels did not select settings. The published energy and STFT thresholds of 9 and 5 serve as fixed transfer baselines.

6.2 Detection and restoration

On 1,728 mixtures from three held-out speakers, event F1 was 0.084 for the original STFT detector, 0.154 after retuning STFT, and 0.475 for the short-context detector. The short-context detector matched 837 events and missed 891, with 960 unmatched predictions. Retuned STFT matched 336 and missed 1392, with 2302 unmatched predictions. On the uncorrupted inputs, short-context detection produced 14 false events and retuned STFT produced 33.

Detector	Precision	Recall	F1	Sample recall	Clean false events
Energy (original threshold)	0.066	0.203	0.100	0.103	70
STFT (original threshold)	0.206	0.053	0.084	0.011	6
STFT (retuned)	0.127	0.194	0.154	0.049	33
Short-context detector	0.466	0.484	0.475	0.212	14

Condition	Original STFT hits	Retuned STFT hits	Short-context hits
-6 dB	6/576	29/576	101/576
+0 dB	10/576	61/576	326/576
+6 dB	75/576	246/576	410/576
silence	24/576	88/576	213/576
speech transient	43/576	146/576	289/576
steady voiced	24/576	102/576	335/576
Unseen real templates	17/648	121/648	293/648

A hit requires interval overlap. Sample recall measures how much of the true event was covered. The strength and location breakdown shows where the detector remains unreliable. Low-level insertions can have less short-window contrast than the surrounding speech. Similar local transients can occur naturally in speech, as the clean control below illustrates. Figure 6 illustrates a matched failure: in BWC arctic a0011, the detector missed the -6 dB keyboard insertion during steady speech but detected the +6 dB version at the same sample position. Even the hit covered only 289 of 381 inserted samples. The unseen-template row pools the three extra ESC-50 recordings, so it supports only a limited check of transfer to new disturbance examples.

Pipeline	Event reduction (dB)	Outside-event RMSE	SI-SDR (dB)	STOI	Median CPU RTF
Unprocessed	0.00	0.000000	28.49	0.985	0.00000
STFT (original)	0.13	0.000120	27.74	0.985	0.00081
STFT (retuned)	0.37	0.000695	25.50	0.984	0.00080
Short-context repair	0.89	0.000500	26.44	0.988	0.00091
GTCRN	6.87	0.001684	23.00	0.974	0.00915

GTCRN achieved 6.87 dB mean event-error reduction, while short-context interpolation achieved 0.89 dB. Their mean outside-event errors were 0.001684 and 0.000500, respectively. Each CPU timing excludes model setup and one warm-up. Clean-region RMSE quantifies waveform alteration. Perceptual quality would require a listening study.

6.3 Manually annotated stops

The clean control checks whether each detector touches unambiguous P, T or K intervals in the corpus's manually checked phones tier. The paired insertion test uses the first eligible token of each phone per utterance and places a 20 ms burst entirely inside the phone interval, near its strongest 2 ms first-difference-energy peak. Tests use three levels per selected token. The peak is an acoustic proxy for the release time. Human annotation identifies the phone interval.

Phone	Original hits	Original clean overlap	Retuned hits	Retuned clean overlap	Short-context hits	Short-context clean overlap
/p/	5/36	0/28	13/36	1/28	27/36	0/28
/t/	0/54	0/32	3/54	5/32	37/54	2/32
/k/	1/42	3/37	12/42	8/37	30/42	1/37

Clean overlap is counted per token, so one false event can touch more than one token. The targeted burst results measure detector sensitivity for a single synthetic disturbance shape and the annotated subset. The utterances repeat across strength variants, and the phoneme counts differ. The experiment does not support a population-level ranking of phonemes.

6.4 Limits and conclusion

The short-context detector improves event F1 over both STFT controls while still missing over half of the test events. Its clean false-event count falls between the two STFT controls, and GTCRN retains a substantial event-restoration advantage. The original OSR experiment established the restoration and preservation trade-off in one recording. The extension repeats that comparison on three unseen speakers and adds a threshold-only control and an annotated consonant test. The sampling of speakers, utterances and event sources remains small. Straight-line repair can remove real speech within the predicted intervals, and partial event coverage can leave interference audible. No formal listening test, streaming latency or peak-memory measurement is included.

7. Reproduction and implementation acknowledgement

Use the README commands to set up dependencies and reproduce both benchmarks. Run scripts/prepare_extended_data.py and scripts/run_extended_experiment.py for the speaker-disjoint evaluation before rebuilding the deliverables. The run creates manifest intervals, frozen thresholds, per-case detection/restoration results, no-op tests, figures and exact example arrays. scripts/export_case.py exports any controlled input; scripts/enhance.py processes a user-supplied WAV without reference labels. See README for platform-specific environment and website instructions.

AI assistance was used in code implementation and documentation. Reported measurements were produced by executing the accompanying experiment code, not taken from expected outcomes or substituted from published GTCRN benchmark numbers.

References

1. S. V. Vaseghi and P. J. W. Rayner, "Detection and suppression of impulsive noise in speech communication systems," IEE Proceedings I, vol. 137, no. 1, pp. 38-46, 1990.

2. R. C. Nongpiur and D. J. Shpak, "Impulse-noise suppression in speech using the stationary wavelet transform," JASA, vol. 133, no. 2, pp. 866-879, 2013. https://doi.org/10.1121/1.4773264

3. C. Medina, R. Coelho and L. Zao, "Impulsive noise detection for speech enhancement in HHT domain," IEEE/ACM TASLP, vol. 29, pp. 2244-2253, 2021. https://doi.org/10.1109/TASLP.2021.3086131

4. X. Rong, T. Sun, X. Zhang, Y. Hu, C. Zhu and J. Lu, "GTCRN: A Speech Enhancement Model Requiring Ultralow Computational Resources," ICASSP 2024. https://doi.org/10.1109/ICASSP48485.2024.10448310. Official implementation: https://github.com/Xiaobin-Rong/gtcrn

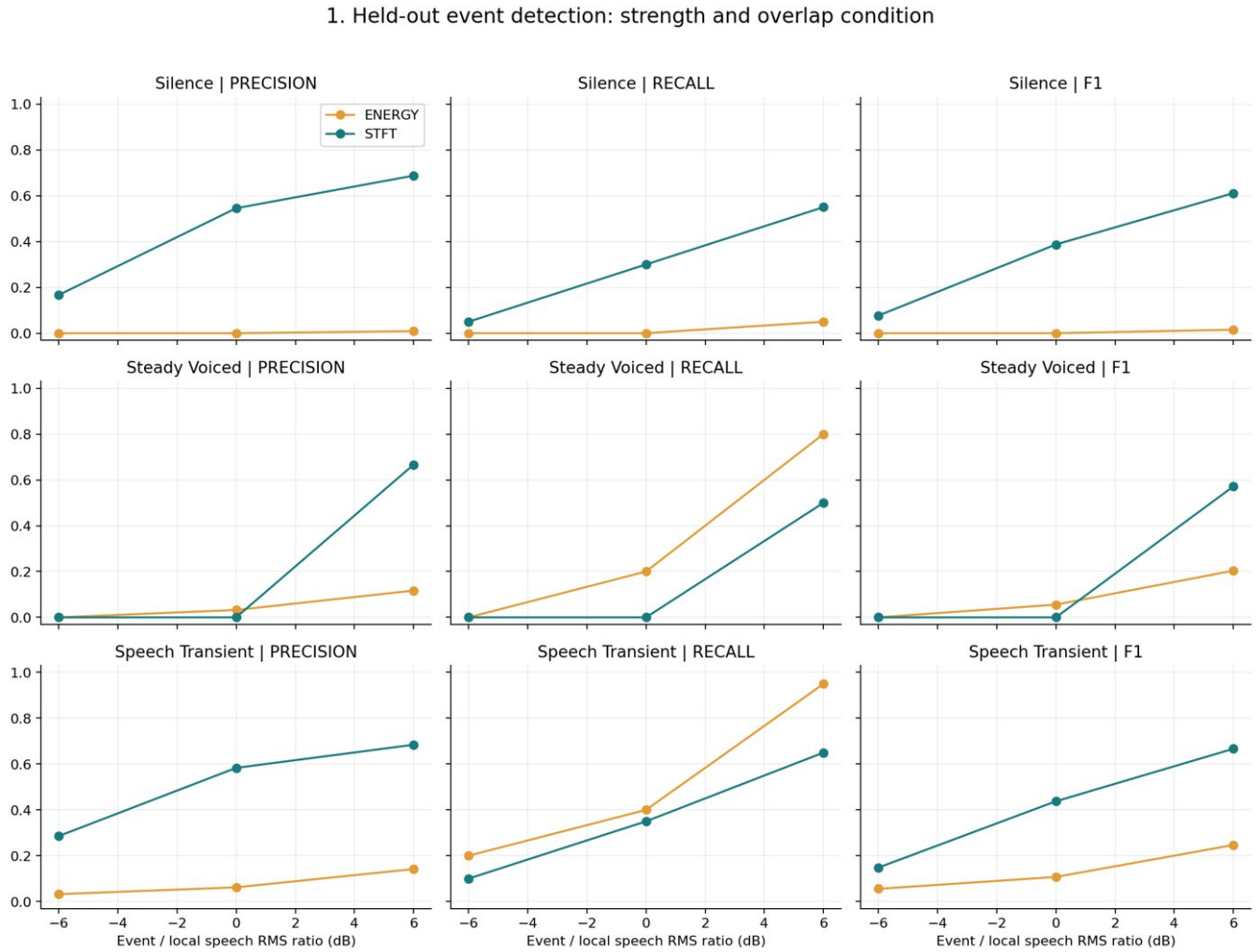
5. L. Zhou, H. Liu, L. Gan, Y. Zhou, M. Niedzwiecki and T.-K. Truong, "A novel sparse adaptive filter for suppressing impulsive disturbance in audio signals," Signal Processing, vol. 241, article 110390, April 2026. https://doi.org/10.1016/j.sigpro.2025.110390. Author code/data: https://github.com/minikatty/Lq_JSLMP

6. K. J. Piczak, "ESC: Dataset for Environmental Sound Classification," ACM Multimedia, 2015. Dataset and attribution: https://github.com/karolpiczak/ESC-50

7. C. H. Taal et al., "An Algorithm for Intelligibility Prediction of Time-Frequency Weighted Noisy Speech," IEEE TASLP, 2011. STOI implementation: https://github.com/mpariente/pystoi

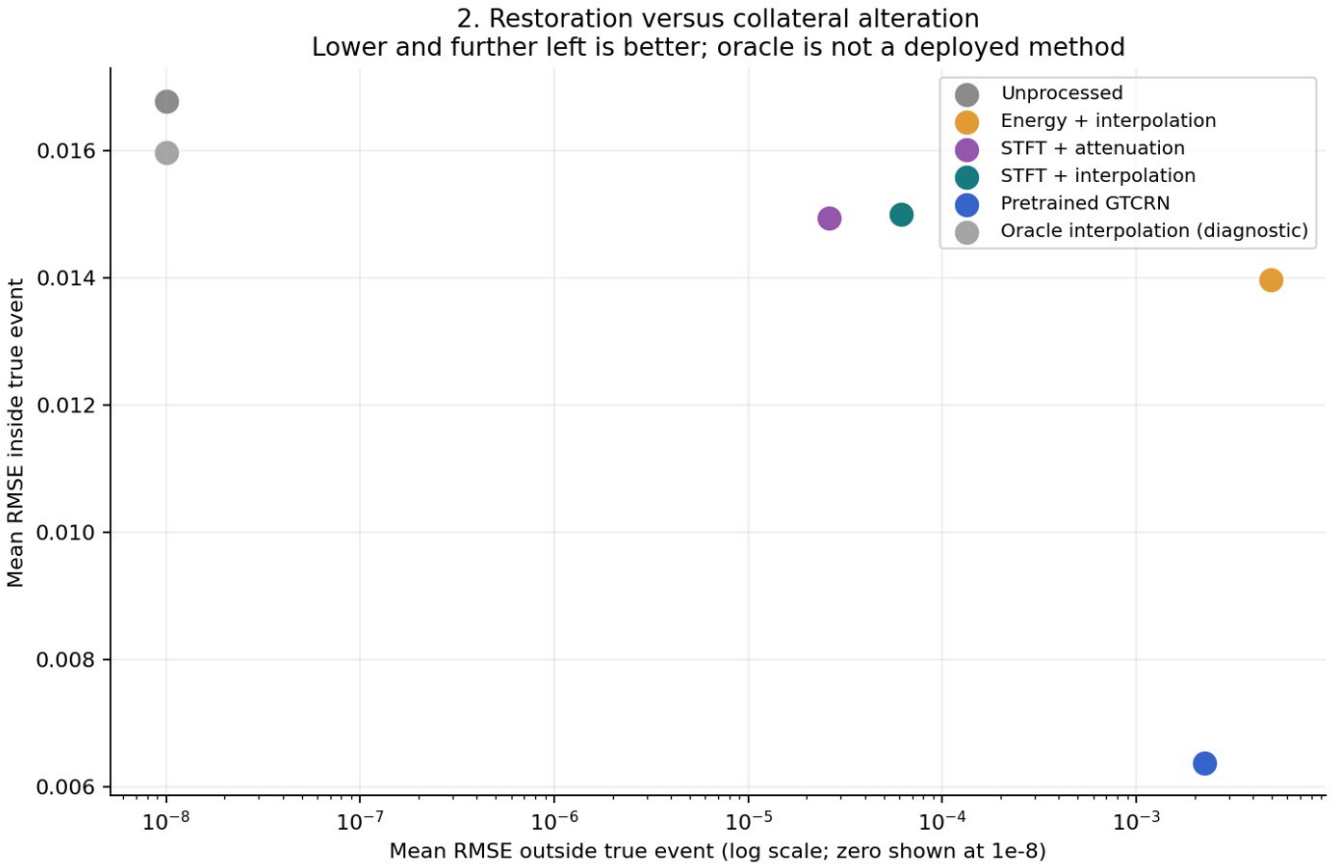
8. G. Zhao et al., "L2-ARCTIC: A Non-native English Speech Corpus," Interspeech 2018, pp. 2783-2787. https://doi.org/10.21437/Interspeech.2018-1110

Figure 1. Detection by strength and acoustic location



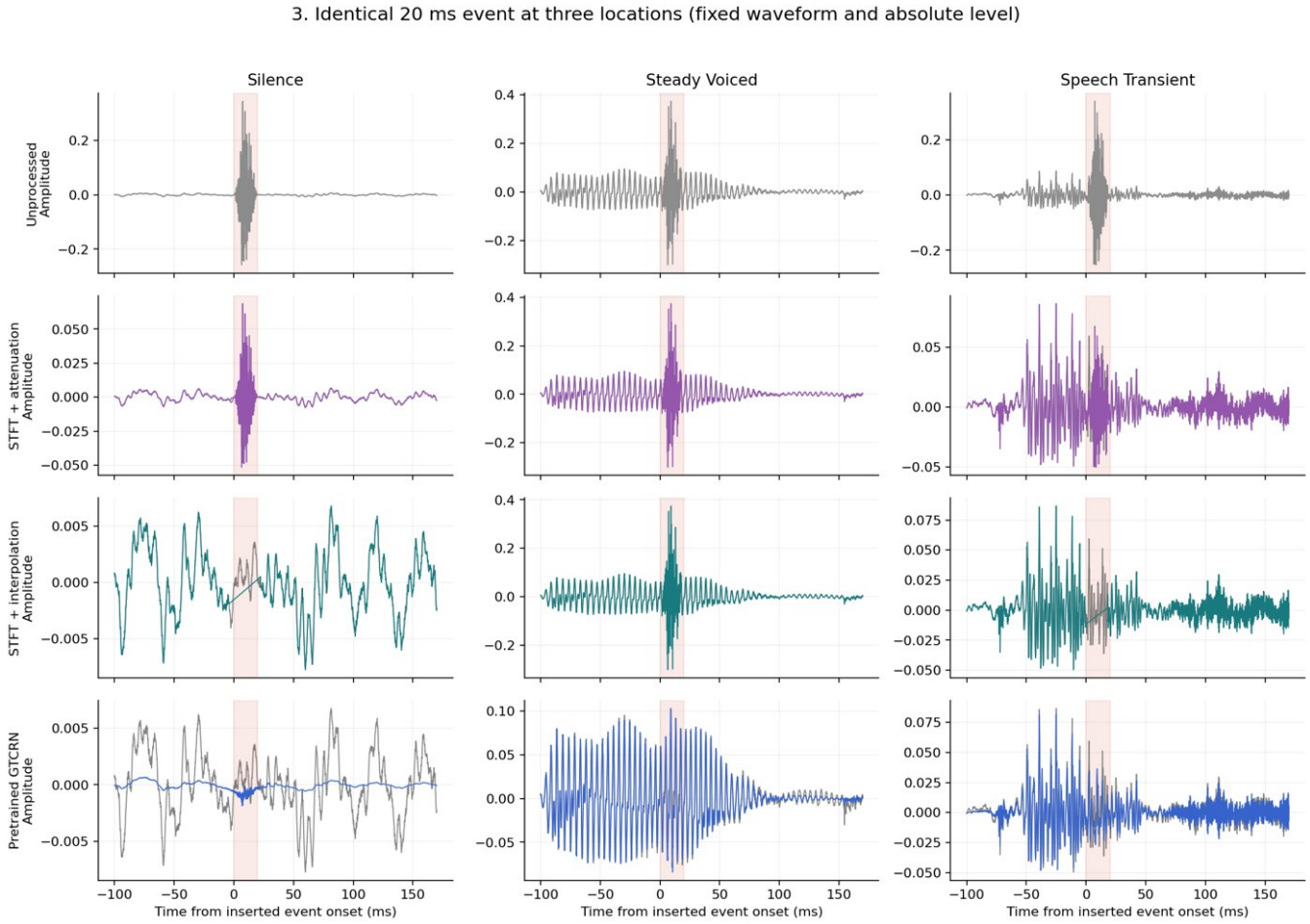
Held-out event metrics; all event templates pooled within each overlap/strength condition.

Figure 2. Event restoration versus collateral alteration



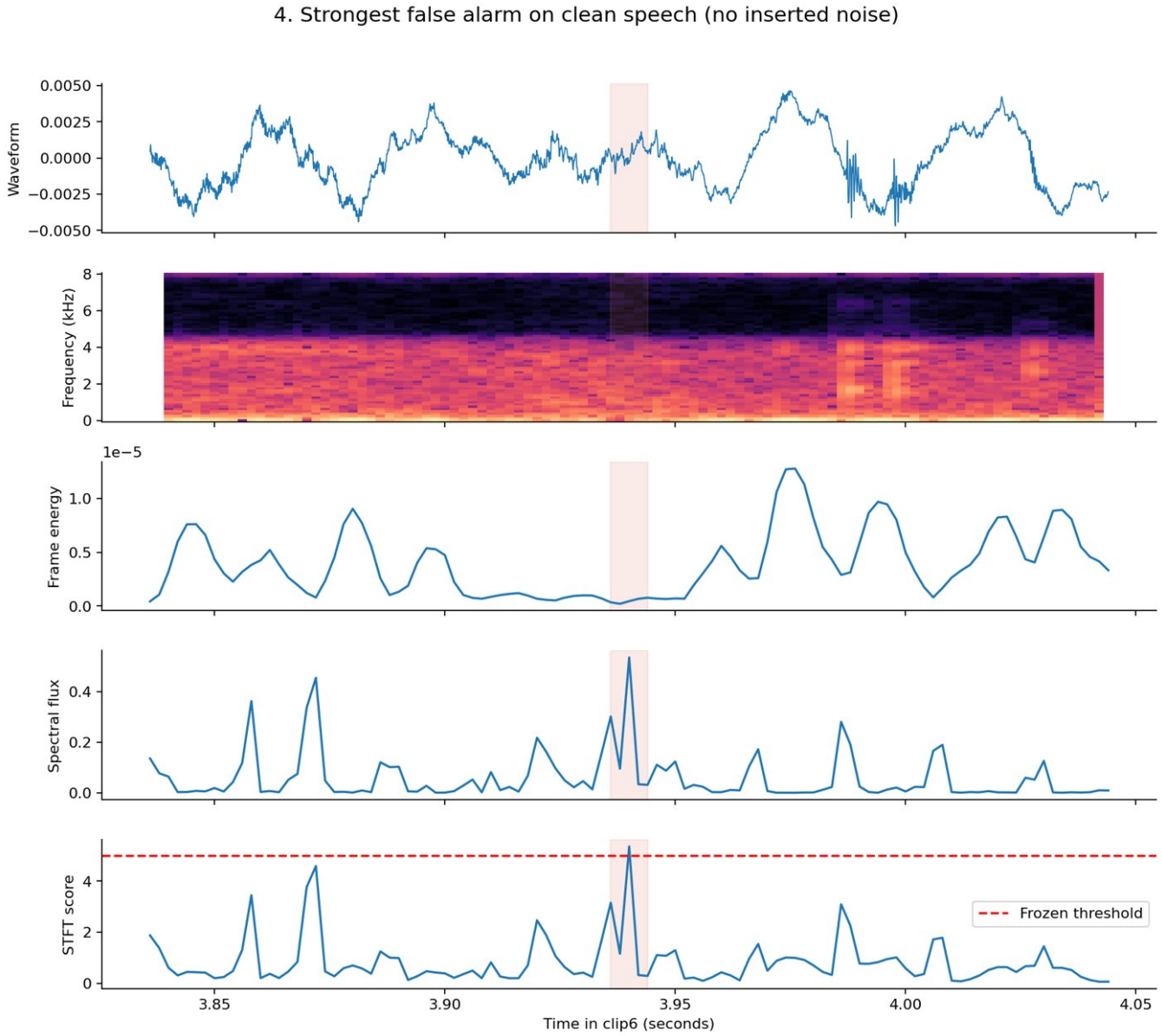
Lower error on both axes is desirable. True-mask interpolation is diagnostic only.

Figure 3. Identical event at different locations



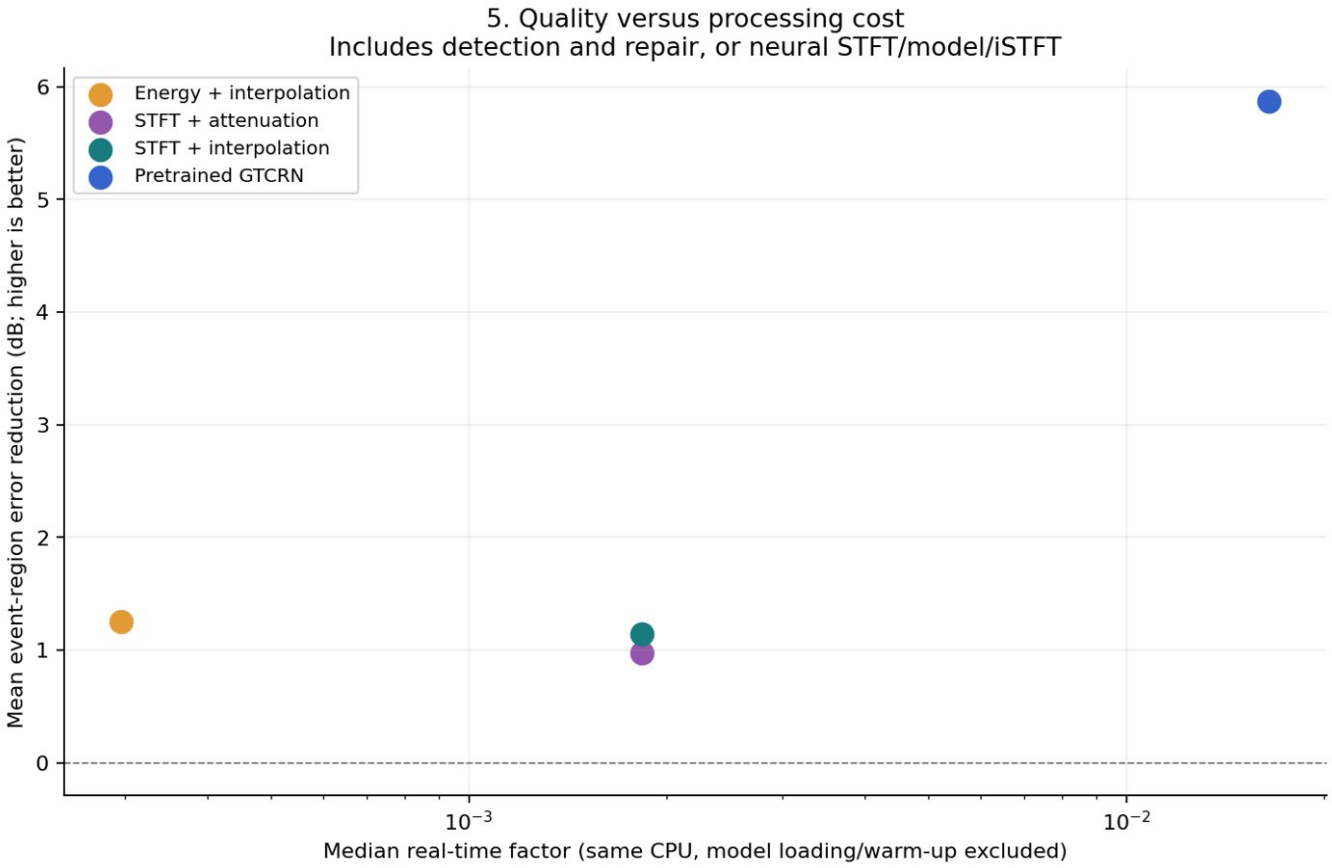
A fixed 20 ms event waveform and absolute amplitude are used for all three insertions.

Figure 4. False detection on clean reference speech



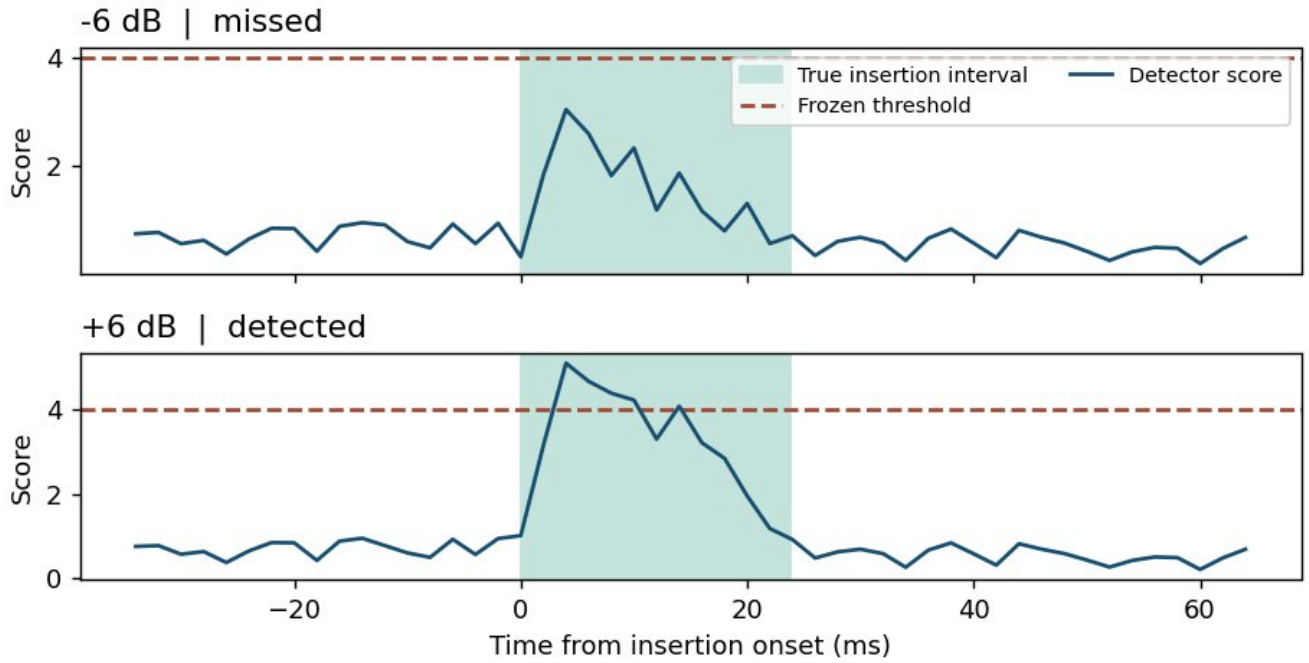
The high normalised flux peak triggers an event despite low absolute energy. No phoneme label is available.

Figure 5. Restoration quality and CPU runtime



Processing is timed on the same CPU. Model loading and warm-up are excluded; no causal latency is inferred.

Figure 6. Matched weak and strong keyboard insertions



Same unseen recording, event template and insertion interval. The score crosses the threshold for the stronger event only.